

ROBUST AUTONOMOUS DRIVING

Evaluating Vision Robustness Under Real-World Image Distortions

Lane Detection • Feature Matching • Vehicle Detection

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Why Robustness Matters

Classical computer-vision and deep-learning methods used in autonomous driving are normally evaluated on **clean** images. On the road, cameras see **low light, motion blur, and rain** every day.

This project measures how much performance each method loses as distortion severity increases — and how much of it can be recovered.

1

Measure degradation

Sweep 3 realistic distortions across 9 severity levels, scored vs. achieved SNR.

2

Enhancement recovery

Restore the image first (denoise / deblur / de-rain), then re-run the same method.

3

Fine-tuning recovery

Adapt the model itself to distorted inputs, using real ground truth.

BDD100K — Real Driving Scenes

Berkeley DeepDrive: large-scale, diverse road-scene images/video with object, lane, and drivable-area annotations — the source for all three tasks.



Dataset 1 — 300 images → Task 1 (Lane Detection) & Task 3 (Vehicle Detection)



Dataset 2 — 178 consecutive-frame pairs from one driving video → Task 2 (Feature Matching)

One Pipeline, Three Perception Methods

Task 1

Lane Detection

Canny + Hough Transform

Classical — Low Level

Task 2

Feature Matching

ORB + BFMatcher + RANSAC

Classical — High Level

Task 3

Vehicle Detection

YOLOv8n

Deep Learning

TASK 1

Lane Detection

Grayscale + blur → Canny edges → trapezoidal ROI mask → morphological closing → probabilistic Hough transform → slope filtering → outlier rejection → weighted regression per side.

Identifies the ego-vehicle's lane boundaries — **a core low-level building block for lane-keeping and path planning.**



Clean-image result

TASK 2

Feature Matching

ORB keypoint detection → BFMatcher (Hamming distance) k-NN → Lowe's ratio test → RANSAC homography, keeping only geometrically consistent inlier matches.

Building block behind visual odometry and tracking — matched between **two consecutive frames of the same driving sequence**, mimicking real frame-to-frame matching.



Clean-image match visualization — two consecutive frames

TASK 3

Vehicle Detection

YOLOv8n (nano) — pretrained on COCO, later fine-tuned on distorted driving images. Locates and classifies car / truck / bus / motorcycle / bicycle.

★ The only task with **real BDD100K ground truth** — unlike Tasks 1 & 2, which use pseudo-GT (clean-image output as reference).



Clean-image detection result

One Headline Metric per Task

Chosen for actually telling a robustness story — monotonic with severity, not flat or noisy.

Task 1

survival_rate

Lane line neither lost nor displaced >45px vs. clean-image reference

1.00

clean baseline

Task 2

match_ratio_vs_baseline

Match coverage relative to the clean-baseline's fixed keypoint budget

0.083

mean, 178 pairs

Task 3

matched_recall

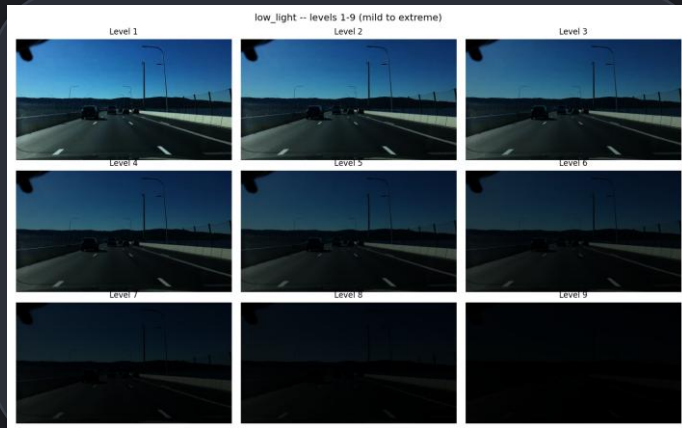
Fraction of real GT boxes found (class-aware IoU ≥ 0.5)

0.366

mean, 300 images, vs. real GT

Three Distortions × Nine Severity Levels

Each swept mild → extreme, scored against achieved SNR (dB).



Low Light

Dusk / night / tunnel driving — brightness + gamma darkening.



Motion Blur

Camera/vehicle motion during exposure — linear blur kernel.



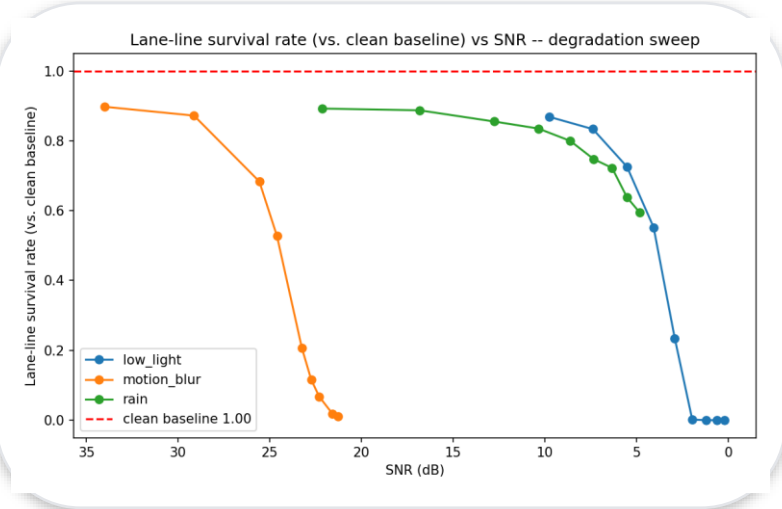
Rain

Rain-streak occlusion — adds clutter, keeps scene structure.

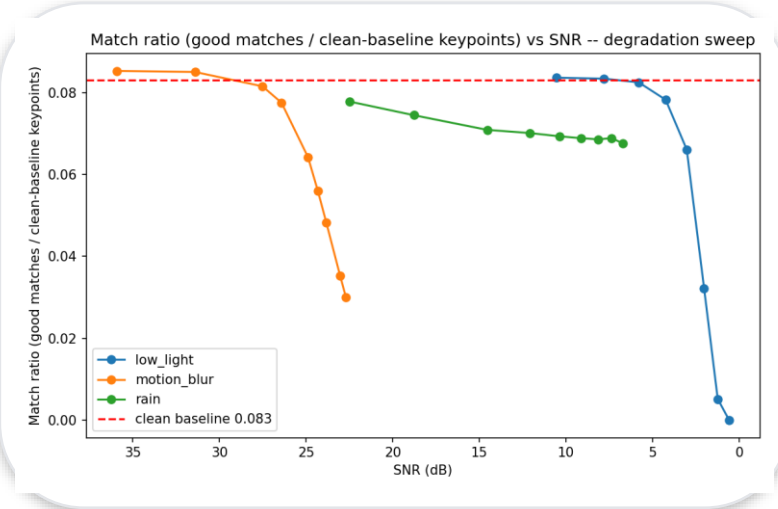
Robustness Falls — At Different Rates

Metric vs. SNR for each task, all three distortions overlaid.

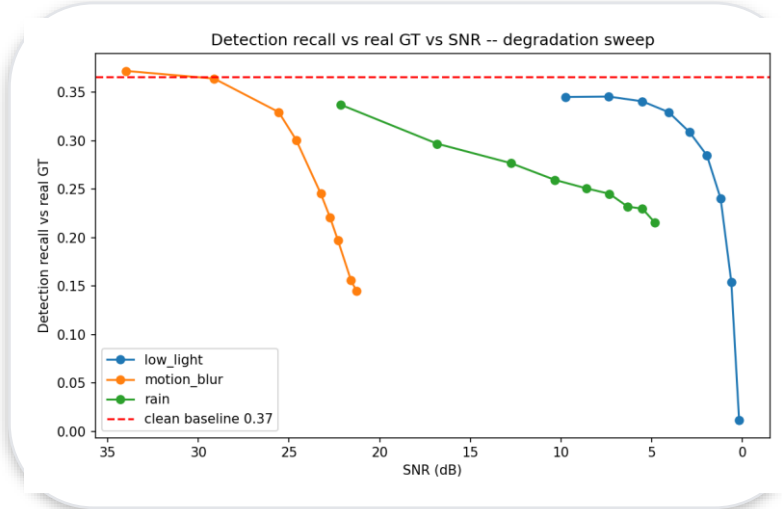
Task 1 — survival_rate



Task 2 — match_ratio_vs_baseline



Task 3 — matched_recall



Task 1: low_light/motion_blur collapse at extreme severity; rain degrades gently. **Task 2:** monotonic decline, sharpest for blur/darkness. **Task 3:** recall drops monotonically; low_light hurts most.

Image Enhancement

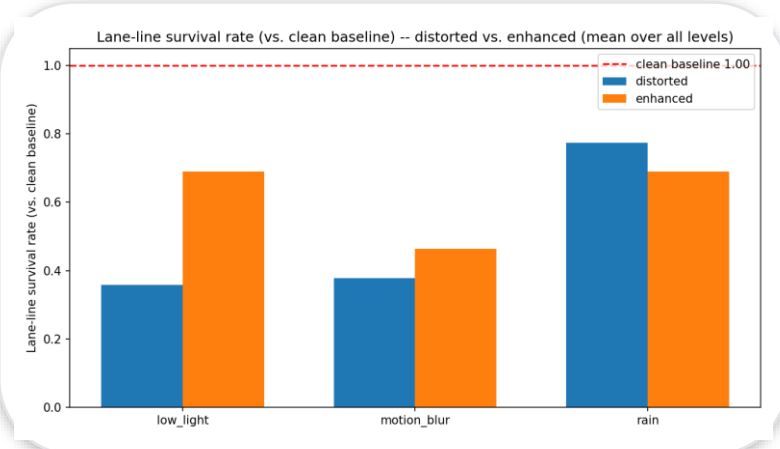
Fixed pre-processing restoration, applied before re-running each method.

Low Light Gamma lift + CLAHE (LAB L-channel)

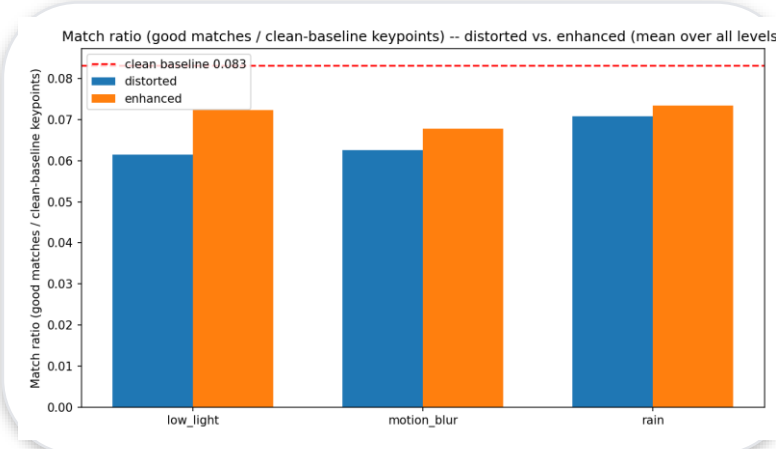
Motion Blur Unsharp masking

Rain Edge-preserving bilateral filter

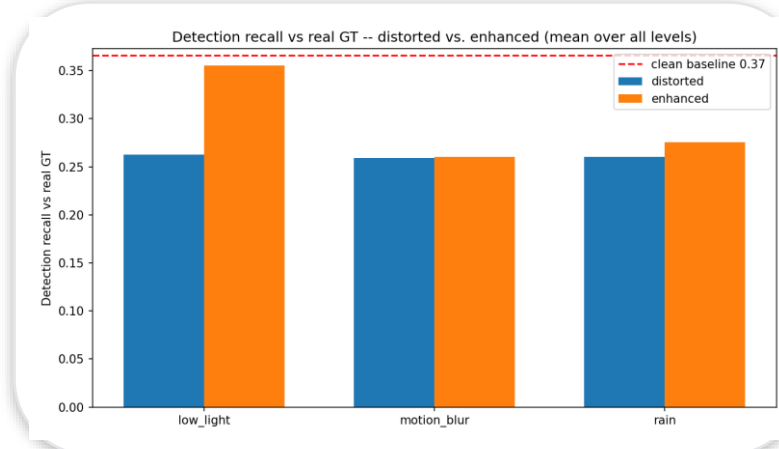
Task 1 — clean vs. distorted vs. enhanced



Task 2 — clean vs. distorted vs. enhanced



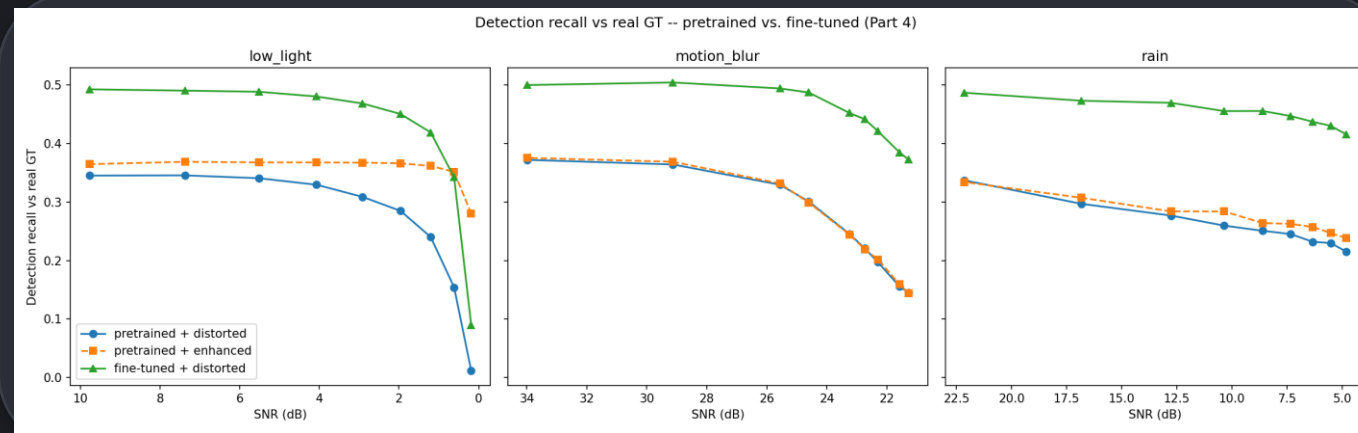
Task 3 — clean vs. distorted vs. enhanced



Enhancement helps unevenly: Task 1's low_light/motion_blur improve but rain slightly worsens (de-rain filter smooths the thin edges Hough needs); Task 2 improves for motion_blur/rain; Task 3 improves for low_light/rain, flat for motion_blur.

Fine-Tuning YOLOv8n

Adapting the model's weights to the distorted domain, using real ground truth.



Mean recall across all 9 levels

	distorted	enhanced	fine-tuned
<i>low_light</i>	0.262	0.355	0.414
<i>motion_blur</i>	0.259	0.261	0.451
<i>rain</i>	0.260	0.275	0.452

Fine-tuning wins on all 3 distortions — the gap over enhancement is largest exactly where enhancement helped least (motion_blur, rain).

Known Issues & Limitations

- **Pseudo-GT for Tasks 1 & 2**

BDD100K's lane-marking and fine-grained-matching label releases weren't locally available; clean-image output serves as the reference instead of human-labeled GT.

- **Dataset pruning**

Task 3 narrowed 675→300 images to match Task 1 and guarantee 100% real-GT coverage; this subset has zero motorcycle/bicycle instances. 105 source images were found corrupt and replaced before any results were produced.

- **CPU-only fine-tuning**

15 epochs, 240 training images, one fixed severity/distortion each — a larger, more varied set and GPU training would likely improve results further.

- **SNR as a proxy**

A uniform-noise measure that doesn't perfectly reflect perceptual severity for structured distortions like rain.

Key Takeaway

**Enhancement recovers some performance;
fine-tuning recovers more.**

Every task degrades under real-world distortion — the two recovery strategies are complementary, not interchangeable: enhancement is cheap and model-agnostic, fine-tuning adapts the model itself and wins when the domain shift is large.

Thank you